

Fecal Egg Identification → Antiparasiticide Selection

Quick-reference guide for dogs and cats. Match what you see on the slide to the likely parasite, then the drug(s) of choice.

1. What You See → Parasite → Drug of Choice

What You See on the Slide	Likely Parasite	Drug(s) of Choice	Key Notes
Thick-walled, brown, round-oval; pitted (Toxocara) or smooth (Toxascaris) surface; unembryonated	Roundworms (Toxocara canis/cati, Toxascaris leonina)	Pyrantel pamoate, fenbendazole, or a macrocyclic lactone (milbemycin, moxidectin)	Zoonotic (visceral/ocular larva migrans). Deworm puppies/kittens at 2, 4, 6, 8 weeks.
Thin-shelled, oval, colorless; morula (8–16 cell) stage inside	Hookworms (Ancylostoma spp., Uncinaria stenocephala)	Pyrantel, fenbendazole, milbemycin, or moxidectin	Zoonotic (cutaneous larva migrans). Benzimidazole/macrocyclic-lactone resistance in A. caninum is now widespread in the US — recheck fecal 2 weeks post-treatment; consider a combination product (e.g., emodepside + praziquantel) or resistance PCR if treatment fails.
Barrel/lemon-shaped, brown, bipolar plugs at each end	Whipworms (Trichuris vulpis)	Fenbendazole (3–5 day course, repeat in 3 weeks); milbemycin or moxidectin also effective	Pyrantel alone is NOT effective. Shedding is intermittent — a negative float doesn't rule it out; treat empirically if signs fit.
Round, thick striated shell with hexacanth embryo; OR rice-grain proglottids/egg packets on stool/perianal fur	Tapeworms (Dipylidium caninum, Taenia spp., Echinococcus spp.)	Praziquantel (drug of choice) or epsiprantel	Standard flotation often misses eggs — diagnosis is usually visual (proglottids). Dipylidium needs flea control too. Echinococcus eggs look identical to Taenia — zoonotic hydatid risk in endemic areas.
Small oval/round oocyst, smooth thin wall, unsporulated (no internal structure visible fresh)	Coccidia (Cystoisospora/Isospora spp.)	Sulfadimethoxine, ponazuril/toltrazuril, or trimethoprim-sulfa	Protozoa, not helminths — NOT killed by anthelmintics/dewormers. Needs a coccidiostat instead.
Oval cyst, smooth wall, 2–4 nuclei and axonemes visible	Giardia spp.	Fenbendazole (5-day course) ± metronidazole	Bathe pet at end of treatment to remove cysts from haircoat. Zoonotic potential debated but hygiene still warranted.
Larvated egg, or free larvae seen directly on the slide	Strongyloides stercoralis	Fenbendazole or ivermectin	More common in puppies, kennels, or immunosuppressed animals. Zoonotic.
Large, thick smooth shell, oval, already embryonated	Physaloptera spp.	Pyrantel or fenbendazole (often needs repeat/higher dose)	Notoriously hard to clear. Associated with chronic vomiting; adults shed eggs intermittently so flotation often misses it.

What You See on the Slide	Likely Parasite	Drug(s) of Choice	Key Notes
Similar to whipworm egg but bipolar plugs rougher/asymmetric	Capillaria / Eucoleus / Pearsonema spp.	Fenbendazole	Respiratory (Eucoleus boehmi) or bladder (Pearsonema) forms exist — note sample source (feces vs. urine sediment).
Very small, round, unsporulated oocyst (smaller than Cystoisospora)	Toxoplasma gondii (cats)	Not treated with dewormers; clindamycin if clinically ill	Zoonotic — key counseling point for pregnant or immunocompromised household members.

2. Drug Class Cheat Sheet

Drug Class	Covers	Common Products
Benzimidazoles (fenbendazole)	Roundworms, hookworms, whipworms, Giardia, some lungworms — broad and safe	Panacur, Safe-Guard
Pyrantel pamoate	Roundworms, hookworms only — NOT whipworms, NOT tapeworms	Nemex, Strongid
Macrocyclic lactones (milbemycin, moxidectin, ivermectin)	Roundworms, hookworms (label-dependent) — usually paired with heartworm prevention	Interceptor Plus, Sentinel, Advantage Multi, Trifexis, Simparica Trio
Praziquantel / epsiprantel	Tapeworms only	Droncit, component of Drontal Plus / Interceptor Plus
Coccidiostats (sulfadimethoxine, ponazuril/toltrazuril)	Coccidia only — not an anthelmintic	Albon, Marquis, Baycox

3. Parasite Species × Drug Class Matrix

✓ = effective / labeled indication. ✓* = effective but caveat applies (see notes below). — = not effective / not indicated.

Parasite Species	Benzimidazoles	Pyrantel	Macrocyclic Lactones	Prazi-quantel	Coccidi o-stats
Toxocara canis / T. cati (roundworm)	✓	✓	✓	—	—
Toxascaris leonina (roundworm)	✓	✓	✓*	—	—
Ancylostoma caninum (hookworm, dog)	✓*	✓*	✓*	—	—
Ancylostoma tubaeforme (hookworm, cat)	✓	✓	✓	—	—
Uncinaria stenocephala (hookworm)	✓	✓	✓	—	—
Trichuris vulpis (whipworm)	✓	—	✓*	—	—
Dipylidium caninum (flea tapeworm)	—	—	—	✓	—
Taenia spp. (tapeworm)	—	—	—	✓	—

Parasite Species	Benzimidazoles	Pyrantel	Macrocyclic Lactones	Prazi-quantel	Coccidiostats
Echinococcus spp. (tapeworm)	—	—	—	✓*	—
Cystoisospora / Isospora spp. (coccidia)	—	—	—	—	✓
Giardia spp.	✓*	—	—	—	—
Strongyloides stercoralis	✓	—	✓*	—	—
Physaloptera spp.	✓*	✓*	—	—	—
Capillaria / Eucoleus / Pearsonema spp.	✓	—	—	—	—
Toxoplasma gondii (cat)	—	—	—	—	—

Notes on ✓*:

- Toxascaris leonina: macrocyclic lactone efficacy is less consistently labeled than for Toxocara — check product label.
- Ancylostoma caninum: benzimidazole and macrocyclic-lactone resistance is now widespread in U.S. dogs; pyrantel resistance also reported. Recheck fecal 2 weeks post-treatment before assuming efficacy.
- Trichuris vulpis: macrocyclic lactone efficacy applies to milbemycin/moxidectin, not all products in the class (e.g., not ivermectin at standard heartworm-prevention doses).
- Echinococcus spp.: praziquantel is effective, but higher-frequency dosing is often recommended in endemic areas due to zoonotic hydatid disease risk.
- Giardia spp.: fenbendazole is effective but often combined with metronidazole (not one of the 5 classes above) for better clearance.
- Strongyloides stercoralis: macrocyclic lactone efficacy refers to ivermectin specifically, at doses higher than standard heartworm prevention.
- Physaloptera spp.: response to pyrantel/fenbendazole is inconsistent; repeat dosing or a different class is often needed to fully clear infection.

4. Caveats Before You Treat

- Confirm with centrifugal fecal flotation, not gravity/passive flotation — it recovers far more eggs, especially Trichuris and Physaloptera.
- Egg count doesn't equal clinical severity, and a negative float doesn't rule out infection (intermittent shedding, prepatent period, single-sex infections). Antigen or PCR testing can catch what flotation misses, especially for hookworms.
- Hookworm drug resistance (Ancylostoma caninum) is a documented, spreading problem in U.S. dogs, affecting both benzimidazoles and macrocyclic lactones. Any apparent treatment failure warrants a fecal recheck 2 weeks post-treatment before assuming reinfection.
- Tapeworm eggs/proglottids are easy to miss on routine flotation — diagnosis is often made by owners seeing motile segments or by visual exam.
- This is a quick-reference tool only. Always confirm current dosing, label indications, and species-specific safety (e.g., ivermectin sensitivity in certain dog breeds) before treating.

Sources: Companion Animal Parasite Council (capcvet.org) guidelines for Ascarids and Hookworms; Cornell University College of Veterinary Medicine on emerging Ancylostoma caninum drug resistance.